BALA SREERANGAM

Cloud Platform Engineer | Cloud Infrastructure, DevOps, Release Engineering, and Site Reliability
Sreerangambala@gmail.com | linkedin.com/in/shanmukh-sreerangam-447529214
shanmukh0410.github.io/portfolio | College Park, MD | Open to relocation

PROFESSIONAL SUMMARY

Cloud Platform Engineer with 4 years operating production AWS at scale across 30+ accounts, 50+ environments, and a multi-tenant SaaS platform serving 20+ enterprise customers in aviation, retail, hospitality, and telecommunications. Introduced Blue/Green RDS deployments that cut deployment time roughly 80% and rebuilt alerting that had reached 50 to 60 pages a day down to 10 to 15 actionable alerts. Builds Terraform modules, Python and boto3 automation, and CI/CD pipelines that remove manual steps from release and provisioning. AWS Certified DevOps Engineer Professional and Security Specialty; completing an M.Eng. in Cloud Computing at the University of Maryland.

TECHNICAL SKILLS

Cloud: AWS (EC2, ECS/Fargate, Lambda, IAM, Organizations, STS, VPC, Route 53, S3, RDS, DynamoDB, ElastiCache, ECR, API Gateway, Step Functions, CloudWatch, EventBridge, SNS, SQS, KMS, Secrets Manager, CloudFormation and StackSets, Glue, Athena, CloudTrail, Config, GuardDuty, Security Hub, Cost Explorer, Transfer Family), GCP and Azure (foundational)

Infrastructure as Code and CI/CD: Terraform (reusable modules, version upgrades, remote state), Atlantis, CloudFormation StackSets, GitHub Actions, CircleCI, Buildkite, Jenkins, Docker, Amazon ECR, multi stage builds, semantic versioning, blue green and rolling deployments, Kubernetes

Languages: Python (boto3), SQL, Bash, Go, JavaScript, TypeScript, Java, C

Security and Observability: IAM least privilege, cross account roles, WAF, Shield, Firewall Manager, GuardDuty, Security Hub, Config, CloudTrail, KMS, secret leak scanning in CI, CIS AWS Foundations, Jamf, Datadog, CloudWatch, Grafana, monitor and threshold design, alert noise reduction, on call

Data and Platform: AWS DMS, Glue, Airflow, Redshift, Athena, QuickSight, S3 data lakes, Snowflake, PostgreSQL, Redis, ETL design, multi account AWS, landing zones, multi tenant SaaS, microservices, serverless, event driven architecture, high availability, disaster recovery, Chef, FinOps and rightsizing

PROFESSIONAL EXPERIENCE

DevOps / Cloud Engineer | LJI

Jan 2022 - Jul 2025

- Introduced **Blue/Green RDS deployments**, cutting deployment time roughly **80%** by building the upgraded instance alongside the live one and switching over in seconds, then paired it with pre and post migration sequencing on a secondary ECS task to reach **zero downtime releases** on any release without an engine upgrade.
- Rebuilt Datadog monitoring from **50 to 60 pages a day at roughly a 10% true positive rate down to 10 to 15 actionable alerts**, a **75 to 80% cut in alert volume**, by observing real spike behavior and setting thresholds per environment and per microservice rather than globally.
- Cut CI security scanning and release tag creation time **60 to 70%** by parallelizing scan jobs, and added secret leak detection so credentials were caught at commit time rather than in an audit.
- Planned and executed **30+ production deployments a week**, including major version upgrades and database migrations, sizing each window from release notes, lower environment dry runs, and client traffic curves in Datadog.
- Eliminated developer environment drift by building a **self service deployment endpoint** on AWS Lambda and API Gateway backed by a Python and boto3 service map, so developers deployed by submitting a service, environment, and tag instead of hand editing ECS, Lambda, and RDS configuration.
- Automated multi step **tenant cloning with Step Functions** across RDS, S3, and DynamoDB, followed by DMS to S3 and an Airflow DAG resync, replacing a long manual runbook for the 5 to 6 clones QA and demo environments needed each week.
- Built an **organization wide audit collector** in Python covering 50+ AWS services in every account, writing to a hierarchical S3 layout in a central account, queried through Athena with email alerting on findings needing immediate action.
- Refactored the Terraform estate into **reusable DRY modules** spanning EC2, ECS, Lambda, VPC, RDS, DynamoDB, S3, CloudFront, API Gateway, and load balancers, and co designed a module version upgrade path so live environments could advance without being rebuilt. Implemented **Atlantis** so plan and apply ran through pull requests, giving every change a reviewable trail.
- Owned the reporting data pipeline, running AWS DMS full load and ongoing replication out of RDS into S3 staging, Airflow loads into Redshift, then Athena and QuickSight with role scoped dashboard access.

- Automated CloudFront provisioning for client websites, migrated AWS Transfer SFTP servers from hostname to static IP with no file loss, and hardened production with **WAF, Shield, Firewall Manager, Security Hub, and GuardDuty**, including web ACLs enforcing client supplied IP allowlists.

Graduate Teaching Assistant | University of Maryland
Aug 2026 - Present | MSML 650, Cloud Computing

- Guide graduate students through hands on labs covering AWS infrastructure, virtualization, containerization, cloud networking, storage, identity management, and scalable application deployment.
- Hold weekly office hours, troubleshoot student cloud deployments, grade technical assignments, and mentor on cloud native development, distributed systems, and infrastructure automation.

Cloud Infrastructure and DevOps Engineer | Quanticore (My Behavior Bridge)
May 2026 - Aug 2026

- Built the organization's **first Infrastructure as Code** in Terraform, with reusable modules for VPC networking, subnets, route tables, NAT and internet gateways, security groups, IAM, EC2, ECS, load balancing, auto scaling, ECR, and DNS, replacing manual AWS console provisioning entirely.
- Designed the **environment strategy**, proposing and standing up a dedicated QA tier between development and production so releases had somewhere to be validated end to end before customers saw them.
- Built the **first CI/CD pipelines in GitHub Actions**, covering source triggered workflows, security checks during build, automated Docker image creation, ECR publishing, and release tags, then automated tenant provisioning and established the monitoring, networking, and IAM practices the platform runs on.

Student Staff | University of Maryland, Accessibility and Disability Service
Feb 2026 - May 2026

- Coordinated accommodated exam operations and scheduling workflows, and built a reusable web scraping tool for instructor email collection plus a Google Sheets scheduling application with constraint based assignment logic and manual override.

PROJECTS

CloudForge | Terraform, Terragrunt, GitHub Actions, VPC, IAM, Route 53, ACM

- Built a landing zone platform that provisions a complete AWS environment from scratch in under 20 minutes, including isolated VPCs, subnets, NAT gateways, security groups, and IAM, with a validate, plan, approve, and deploy pipeline plus remote state locking.

AutoOps | GitHub Actions, Docker, ECR, ECS, Terraform, Python

- Built a CI/CD platform carrying source to production with no manual step, covering security scanning, multi stage Docker builds, semantic versioning, ECR publishing, rollback, and post deploy validation.

Sentinel and InfraGuard | Lambda, EventBridge, SNS, CloudTrail, AWS Config, Python

- Built an event driven governance platform that continuously scans QA, UAT, and production for configuration drift and security issues across IAM, public exposure, encryption, and logging, auto remediates critical findings, and centralizes reporting.

RESEARCH

Stratosyn, Intelligent Cloud Governance and Autonomous Infrastructure Platform | Faculty advised, Sep 2025 - Present

- Built a multi account discovery engine using AWS Organizations enumeration, cross account role assumption, and concurrent multi region scanning, solving sparse schema normalization so the schema for a service becomes the union of every key seen anywhere in the organization, null filled where a resource omits it, stored as partitioned NDJSON in S3 with Glue and Athena providing schema on read.
- Researching automated compliance validation against CIS AWS Foundations, NIST, and SOC 2, blast radius graph generation, and safe remediation with impact analysis, rollback, and audit trails.

CERTIFICATIONS

AWS Certified DevOps Engineer Professional | **AWS Certified Security Specialty** (valid through 2029) | AWS Certified Solutions Architect Associate | AWS Certified Cloud Practitioner | HashiCorp Certified Terraform Associate (003) | Jamf Certified Associate

EDUCATION

Master of Engineering (M.Eng.), Cloud Computing | University of Maryland
Sep 2025 - May 2027 (expected)